

Technische Universität Chemnitz

Exploring Reinforcement learning strategies in automated wheel truing

Master's Thesis

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Abstract

This thesis attempts to find a solution to automatic spoked wheel truing through reinforcement learning in a simulated environment. We test and evaluate a set of different algorithms, state-spaces and reward schemes.

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Chapter 1

Introduction

1.1 Motivation

short motivation preferably relating the problem to material stress optimization problems.

1.2 Wheel truing

describing the problem and mention sim-to-real outlook

1.3 Thesis Structure

Chapter 2

Theory

TODO:

- start algorithm chapters (bullet points)
- finish physics section

OVERVIEW: This chapter should outline the necessary theoretical background on deep reinforcement learning, as well as the relevant physics from Dr. Ford's thesis on bicycle wheel dynamics.

OPEN QUESTIONS:

2.1 Reinforcement learning basics

Here I want to cover the basic ideas, definitions and mathematical prerequisites. This will probably include:

- Markov states
- states, actions and their values
- value iteration and policies
- function approximation and deep learning
- data sampling and replay buffer

After these basics, I want to describe the main ideas and various improvements of the different approaches of the three selected algorithms in the next subsections:

2.2 DeepQ (Rainbow)

2.3 Policy gradients (PPO)

2.4 World-Models (TDMPC)

2.5 Wheel Truing Physics

The environment used in this thesis is built upon a simulation based on a physical model of spoked bicycle wheels presented in [Ford2018ReinventingTW]. This section covers a brief overview of the problem formulation and numerical solution to the spoked wheel system.

2.5.1 Structural components and modeling assumptions

The spoked wheel is essentially built from three different components: the hub, rim and a number of spokes. Relative to the spokes and rim, the hub is a rigid structure of high stiffness; it is therefore modeled as stationary and non-deformable and provides the boundary condition of the spoke-rim system through its diameter d_h and width w_h . The rim is in essence a thin-walled, curved beam with constant radius R measured at the shear center of the cross-section. This cross-section features a complex cross-sectional geometry exhibiting monosymmetric geometry (symmetric about the radial axis but not the lateral axis) which introduces coupling between lateral bending and torsional deformations. To describe these interactions, the rim is modeled as being made from a linearly elastic material obeying the generalized Hooke's law, which forms a beam obeying Euler-Bernoulli theory. This boils down to neglecting shear deformations that are not associated with uniform torsion (plane sections remain plane and perpendicular to the neutral axis during bending) and ignoring in-plane distortion of the rim cross-section. From Hooke's law we need the Young's and shear moduli E and G respectively. The spokes are modeled as ideal, linear elastic bar elements, deforming only along their length with a Young's modulus E_s . The spokes are connected to the rim and hub by idealized, moment-free, ball-and-socket joints, meaning they do not transmit bending moments or torques. To describe deformations in the wheel, a local cylindrical coordinate system is defined through a right-handed orthonormal triad. The basis vectors $\{e_1, e_2, e_3\}$ of this triad point in lateral (out of plane), radial (toward hub) and tangential (increasing arc length) directions. The origin travels with the rim centroid specified by an angular coordinate $\theta \in [-\pi, \pi]$. With this system we can describe the spoke position through θ and directions through direction cosines $\{c_1, c_2, c_3\}$ defined by the spoke unit vector $n = c_1 e_1 + c_2 e_2 + c_3 e_3$ with $c_1^2 + c_2^2 + c_3^2 = 1$.

Chapter 3

Environment

TODO:

- introduce the thinking behind different reward functions
- motivate and list wheel parameters (done)
- make sure the wheel parameters have the right units and descriptions (sorta done needs last look)
- in reward section introduce terminating and penalty condition (done)

OVERVIEW: Here I want to describe the process and considerations taken to transform the wheel simulation into a suitable reinforcement learning environment. Loosely, this will include:

- Isolating and optimizing the necessary physics calculations
- State-space design ([n]rimpoints, spoke tensions and Fourier coefficients)
- Action-space formulation (continuous one-hot, discrete and all spokes)
- Reward function design
- Episode structure and termination conditions

OPEN QUESTIONS:

- should we add a reward function similar to raw as the sum over turns? or even
- a mixed reward function combining spoke turns and rim displacement?

3.1 On Wheel Parameters

From the previous chapter we can infer that we have quite a few open parameters describing the wheel:

Table 3.1: Wheel simulation parameters

Parameter	Value	Description
n_spokes	36	Number of spokes
max_tension_threshold	1200 N	Maximum allowable spoke tension
max_tension_penalty	true	Enable penalty for exceeding max tension
hub_width	50 mm	Hub width (distance between flanges)
hub_diameter	40 mm	Hub flange diameter
rim_radius	300 mm	Rim radius (at shear center)
rim_area	100 mm ²	Rim cross-sectional area
rim_I_lat	1500 mm ⁴	Rim lateral moment of inertia (I_2)
rim_I_rad	3000 mm ⁴	Rim radial moment of inertia (I_1)
rim_J_tor	500 mm ⁴	Rim torsion constant (J)
rim_young_mod	69 GPa	Rim Young's modulus (aluminum)
rim_shear_mod	26 GPa	Rim shear modulus
rim_I_warp	0.0 mm ⁶	Rim warping constant (I_w)
spokes_crossings	3	Number of spoke crossings (lacing pattern)
spokes_diameter	2.0 mm	Spoke wire diameter
spokes_young_mod	210 GPa	Spoke Young's modulus (steel)
number_modes	40	Number of Fourier modes (N)
init_tension	800 N	Initial spoke pretension (T_p)

3.2 On the Starting State

On reset of the environment, i.e., start of a new episode, we need to construct an out-of-true wheel. We do this by starting with a true wheel with completely uniform tensions and then bringing it out of true by choosing n_{choice} spokes and turning them some random uniformly sampled value from $[-\max_{\text{turn}}, \max_{\text{turn}}]$. We choose somewhat arbitrarily $n_{\text{choice}} = 5$ and $\max_{\text{turn}} = 1$.

3.3 On State-Space Formulations

The complete state is defined by the displacement field and spoke tensions in equilibrium as explained in the previous chapter. For representing the state we can choose an arbitrary number of points around the rim in terms of the rotation angle θ . We make this choice to be 360 points, equidistant around the circle. For each of these points, we can simulate the displacement triad, giving us the local lateral, tangential and radial displacement. As it is not realistic to discern the tangential displacement by eye when truing a wheel, or even measure it through sensors, it is omitted from the state. The spoke tensions in equilibrium are calculated by the simulation in units of Newtons. To reduce the variation in absolute values in the state (displacement vectors on the order of a few millimeters vs. spoke tensions on the order of hundreds of Newtons), the tensions are normalized by dividing through the initial tension. Since the displacement around

the wheel is periodic by nature, we can also express this displacement in terms of its Fourier coefficients. For the lateral and radial displacement fields $u_{\text{lat}}(\theta)$ and $u_{\text{rad}}(\theta)$, we compute the discrete Fourier transform coefficients up to N harmonics:

$$c_k = \frac{1}{M} \sum_{j=0}^{M-1} u(\theta_j) e^{-2\pi i k j / M}, \quad k = 0, 1, \dots, N \quad (3.1)$$

where $M = 360$ is the number of sample points, $\theta_j = 2\pi j / M$ are the equidistant angular positions, and $u(\theta_j)$ represents either the lateral or radial displacement at position j . The Fourier representation consists of the real and imaginary parts of these coefficients for both displacement components, yielding a more compact state representation of $4(N + 1)$ features. We choose $N = 30$ by the elbow-graph heuristic on the distance of the original displacement points to the reconstructed points from the Fourier coefficients with some margin of safety. To explore how much of the full state is required for the agent to learn and if we could get away without measuring the spokes in a potential physical implementation, we combine these parts into the following state-space options:

- only displacement points
- displacement points and normalized spoke tensions
- Fourier coefficients
- Fourier coefficients and normalized spoke tensions

3.4 On Action-Space Formulations

Since DQN variants only work with discrete and TDMPC2 only with continuous action spaces, we need at least two formulations. Starting with the discrete formulation, we let the agent choose from $[2 \cdot n_{\text{spokes}}]$ actions corresponding to a 0.1 turn in $+/-$ direction of the corresponding spoke. We can transform this directly into a 2D continuous function by one-hot encoding the spoke vector and letting the second vector control the adjustment size. Additionally, we can also formulate a continuous action by letting the agent adjust all the spokes at once with a vector of length n_{spokes} , giving us the following three action spaces $\mathcal{A}$:

- **Continuous spoke selection:** two-dimensional continuous space

$$\mathcal{A}_{\text{continuous}} = [0, n_{\text{spokes}} - 1] \times [-1, 1] \subset \mathbb{R}^2 \quad (3.2)$$

where n_s is the number of spokes. An action $a = (s, \Delta t) \in \mathcal{A}_{\text{continuous}}$ specifies a continuous spoke index s and a tension adjustment Δt .

- **Discrete spoke selection:**

$$\mathcal{A}_{\text{discrete}} = \{0, 1, 2, \dots, 71\} \quad (3.3)$$

representing 72 distinct action choices, where each action corresponds to a specific spoke-tension adjustment pair.

- **All spokes adjustment:** n_{spokes} -dimensional continuous space

$$\mathcal{A}_{\text{all_spokes}} = [-1, 1]^{n_{\text{spokes}}} \subset \mathbb{R}^{n_s} \quad (3.4)$$

where an action $a = (\Delta t_1, \Delta t_2, \dots, \Delta t_{n_s}) \in \mathcal{A}_{\text{all_spokes}}$ specifies simultaneous tension adjustments for all spokes.

3.5 On Reward Functions

The formulation of the reward is notoriously important for successful reinforcement learning. (citations?)

We can evaluate the current state of the wheel by either looking at the remaining spoke turns from its initial true state or by measuring the total displacement by summing over the displacement vectors weighted by $d\theta$:

$$\|s_t\| \approx \sqrt{\sum_{k=1}^{N-1} \frac{\Delta\theta_k}{2} [\|u(\theta_k)\|_2^2 + \|u(\theta_{k+1})\|_2^2]} \quad (3.5)$$

where $\|u(\theta)\|_2^2 = \sum_{d=1}^D u_d^2(\theta)$ is the squared Euclidean norm of the displacement vector at angle θ , and define the step-improvement I_{step} as $\|s_{t-1}\| - \|s_t\|$.

- **Raw reward:** Directly penalizes the remaining displacement in the rim

$$r_t^{\text{raw}} = -\|s_t\| \quad (3.6)$$

- **Normalized reward:** Provides binary feedback based on step-wise improvement

$$r_t^{\text{normalized}} = \begin{cases} 0 & \text{if } I_{\text{step}} > 0 \\ -1 & \text{if } I_{\text{step}} \leq 0 \end{cases} \quad (3.7)$$

- **Spokes reward:** Provides binary feedback based on spoke adjustment magnitude

$$r_t^{\text{spokes}} = \begin{cases} 0 & \text{if } |\Delta t_i^{(t)}| \leq |\Delta t_i^{(t-1)}| \text{ for all } i \\ -1 & \text{otherwise} \end{cases} \quad (3.8)$$

where $\Delta t_i^{(t)}$ denotes the turns applied to spoke i at time step t .

Additionally, we introduce two stopping conditions to reward the agent for producing a true wheel or punish for exceeding a tension threshold. Some research into actual manual wheel truing shows that the "trueness" is often evaluated only by the rim displacement with less concern for the uniformity of spoke tensions. The state of a true wheel is often seen as dependent on the material quality of the wheel. In this context, we often see discussion around the concepts of "decent," "good," and "golden" wheels, where these generally have the following specifications:

- A decent wheel refers to a common quality wheel which is considered reasonably true with a maximum rim displacement of 1–2 mm and a relative tension deviation of 15–20%
- A good wheel refers to a good quality wheel with a maximum rim displacement of 0.5–1 mm and a relative tension deviation of 5–10%
- A golden wheel refers to an excellent quality wheel with a maximum rim displacement of less than 0.5 mm and a relative tension deviation of under 5%

Since in simulation we can assume no errors or inherent tension in the material of the rim, we can orient ourselves close to the golden standard. From this we introduce the success condition as:

$$|u_d(\theta_i)| < 0.3 \text{ mm} \quad \forall d \in \{1, \dots, D\}, \forall i \in \{1, \dots, N\} \quad (3.9)$$

$$700 \text{ N} \leq T_j \leq 900 \text{ N} \quad \forall j \in \{1, \dots, n_s\} \quad (3.10)$$

$$\max_j T_j - \min_j T_j \leq 0.1 \cdot \bar{T} \quad (3.11)$$

where $u_d(\theta_i)$ represents the displacement in dimension d at discrete angle θ_i , T_j is the tension in spoke j , and $\bar{T} = \frac{1}{n_s} \sum_{j=1}^{n_s} T_j$ is the mean spoke tension. When these conditions are met, the episode terminates with a bonus reward of $r_t = 20$.

For the penalty condition we consider the fact that most wheel manufacturers give a maximum tension of 120 kg force on a spoke. We consider the episode as failed when a single spoke exceeds this threshold and we punish the agent with a reward of $r_t = -20$. To avoid filling the replay buffer with exceptionally bad states at the beginning, we also terminate the episode after exceeding this threshold. So our second stopping condition can be formulated as:

$$\text{if } \exists j \in \{1, \dots, n_s\} : T_j > T_{\max} \quad \text{then} \quad r_t \leftarrow r_t - 20, \text{ truncated} = \text{True} \quad (3.12)$$

where $T_{\max}$ is the maximum tension threshold.

Chapter 4

Experiments and Methodology

Roughly, a short description of the experimental design would be as follows:

I plan to do two phases of experiments. In phase 1, the goal is to establish a baseline for the different algorithms. I do this by running single training experiments on all the combination of state-, action and reward schemes as well as two network sizes. (maybe without tan displacement and with one state with reduced point numbers)

TODO: justify the parameters used here for each algorithm

- here list the problems (some experiments done with insufficient logging and states i dont want to keep as well as non integrated penalization and maybe bad stopping condition)

TODO: look through the experiments already done and list what to keep and what needs to be redone.

Phase 2 should include a hyper-parameter optimization of the most promising candidates from phase 1.

In this chapter I also want to talk about the evaluation metrics.

TODO: is there something else i want to look to better evaluate?

Chapter 5

Results and Analysis

5.1 Baseline Performance

- performance of the algorithms in phase 1
- what can we say about the influence of reward functions and
- state space
- action space
- cross factor analysis

5.2 hyper-parameter optimization

I am not sure yet which candidates will do well and what exactly I want to evaluate here after optimizing the best candidates.

Chapter 6

Discussion and Conclusion

Appendix

.1 Additional Figures

Additional figures that didn't fit in the main chapters.

.2 Additional Tables

Additional tables with detailed data.

.3 Code Listings

Extended code listings that are too long for the main text.

.4 Experimental Details

Detailed experimental configurations and parameters.